

## Problem Set 3

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CEE 260/MIE 273: Probability & Statistics in Civil Engineering

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*Due Wednesday, September 30, 2026 at 11:59 PM as PDF uploaded via Gradescope. If it helps and if possible, you can write your responses directly on this document and upload it instead. **Show as much work as possible in order to get FULL credit.** There are 4 problems with a total of 32 points available.*

### Problem 1 (8 points)

Respond “T” (True) or “F” (False) to the following statements. Use the boxes provided. Each response is worth 1 point.

- (i) ☐ Given three events  $D$ ,  $A$  and  $G$ . If  $P(\overline{D}|AG) = 1$ , then  $P(D)$  is an impossible event.
- (ii) ☐ Two events  $A$  and  $B$  can both be mutually exclusive and yet collectively exhaustive.
- (iii) ☐ If  $P(A|B) = P(A)$  for a set of events  $A$  and  $B$ , then both events are dependent.
- (iv) ☐ Events  $E$  and  $F$  are mutually exclusive and collectively exhaustive. If  $P(E) = 0.3$ , then  $P(F) = 0.7$ .
- (v) ☐ Events  $E$  and  $F$  are collectively exhaustive but *not* mutually exclusive. For a given event  $A$ , its probability can be obtained by  $P(A) = P(AE) + P(AF)$ .
- (vi) ☐ The number of ways 6 books can be arranged on a bookshelf is 720. If two of the books are identical, then the total number of distinct arrangements is 360.
- (vii) ☐ The number of distinct subgroups of size  $n$  that can be formed from a larger group of  $m$  objects is given by  $\frac{m!}{n!(n-m)!}$ .
- (viii) ☐ The events  $E_1$  and  $E_2$  are independent. If  $P(E_1) = 0.4$  and  $P(E_1E_2) = 0.04$ , then  $P(E_2) = 0.1$ .

**Problem 2 (9 points)**

The airline industry in a certain country is subject to labor strikes by the pilots (event  $A$ ), mechanics (event  $B$ ) or flight attendants (event  $C$ ). The probability of strikes by each of the individual groups in the next 3 years is given by:

$$P(A) = 0.03 \quad P(B) = 0.05 \quad P(C) = 0.06$$

Assuming that events  $A$ ,  $B$  and  $C$  are statistically independent, find the probability that:

- [1]      **(a)** only pilots will strike in the next 3 years
- [3]      **(b)** both mechanics and flight attendants will strike in the next 3 years
- [5]      **(c)** any 2 of the 3 groups will strike in the next 3 years

**Problem 3 *Bayes' Theorem (8 points)***

Given an earthquake of intensity

$$X : \{\text{light } (L), \text{moderate } (M), \text{important } (I)\} \quad (0.1)$$

and a structure that can be in a state

$$Y : \{\text{damaged}(D), \text{undamaged}(\overline{D})\} \quad (0.2)$$

The likelihood of damage given earthquake intensity is given by the following conditional probabilities:

$$P(D|L) = 0.01$$

$$P(D|M) = 0.10$$

$$P(D|I) = 0.60$$

and the prior probability of each intensity is given by

$$P(L) = 0.90$$

$$P(M) = 0.08$$

$$P(I) = 0.02$$

(a) Draw and label a Venn diagram illustrating the given events.

[5]

(b) Find the total probability  $P(D)$ .

[3]

**Problem 4 *Bayes' Theorem (continued; 7 points)***

Use the quantities provided and the results from Problem to answer the following questions.

[6] (a) Use Bayes' Theorem to find the posterior probabilities  $P(L|D)$ ,  $P(M|D)$  and  $P(I|D)$ .

[1] (b) Show that the probabilities  $P(L|D)$ ,  $P(M|D)$  and  $P(I|D)$  sum up to 1.